

DAVIDE BALDELLI

PhD researcher at Polytechnique Montréal and Mila

✉ davide.baldelli@mila.quebec  [dundalia.github.io](https://github.com/dundalia)  [github](https://github.com)  [linkedin](https://www.linkedin.com/in/davidebaldelli)  [scholar](#)

EDUCATION

MILA and Polytechnique Montréal

Jan 2025 - Current

PhD in Computer Engineering (GPA: 4.0)

- Supervisors: Sarath Chandar, Amal Zouaq.
- Courses included Representation Learning, Reinforcement Learning.

University of Bologna

Sept 2021 - Oct 2023

MSc in Artificial Intelligence (Grade: 110/110 cum laude)

- Courses included Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing.
-  Thesis: A TWO-step LLM Augmented distillation method for passage Reranking.

University of Florence

Sept 2017 - Mar 2021

BSc in Mathematics (Grade: 110/110)

- Courses included Analysis, Linear Algebra, Geometry, Probability and Statistics, Logic and Computability.
-  Thesis: Big Deviations for Markov Chains.

PUBLICATIONS

Probabilistic Calibration Is a Trainable Capability in Language Models

PrePrint

Davide Baldelli, Sruthi Kuriakose, Maryam Hashemzadeh, Amal Zouaq, Sarath Chandar

LLMs Can't Play Hangman: On the Necessity of a Private Working Memory for Language Agents

CoLLAs 2026 (Oral)

MemAgents Workshop @ ICLR 2026

Davide Baldelli, Ali Parviz, Amal Zouaq, Sarath Chandar

CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design

TMLR

Prashant Govindarajan, Davide Baldelli, Jay Pathak, Quentin Fournier, Sarath Chandar

TWOLAR: A TWO-Step LLM-Augmented Distillation Method for Passage Reranking

ECIR 2024 (Oral)

Davide Baldelli, Junfeng Jiang, Akiko Aizawa, Paolo Torrioni

POSITIONS AND PROGRAMS

MATS Fellowship (10.0 + 10.1 Extension)

Jun 2026 - Present

- Supervised by Mirko Bronzi and Damiano Fornasiere from LawZero.
- Investigating introspection capabilities in language models.
- Selected for the MATS 10.1 extension phase (from Sep 2026) to continue the project.

ARENA 7.0 (Alignment Research Engineer Accelerator)

Jan 2026

- Technical AI safety program focused on building ML engineering skills for alignment research.

Loop AI Labs Inc.

Sept 2022 - Dec 2024

- Machine Learning Engineer.
- Co-supervised a master's thesis titled "Self Assessment Tool for Medical Exam Candidates" in collaboration with the University of Padova.

National Institute of Informatics (NII)

Mar 2023 - Aug 2023

- International Internship Program.
- Supervisors: Akiko Aizawa, Paolo Torrioni.
- Developing an LLM based text-reranker.

SERVICE AND LEADERSHIP

INF8245AE Machine Learning – Polytechnique Montréal Teaching Assistant	Fall 2026
Fatima Al-Fihri Predoctoral Fellowship Research Mentor	Fall 2026
Mentoring a fellow on a machine learning research project through the Fatima Institute for Global AI Research, a program supporting prospective PhD applicants from low- and middle-income countries.	
SPAR (Supervised Program for Alignment Research) Research Mentor	Fall 2026
Mentoring two research fellows on building robust activation monitors for language models.	
CoLLAs 2026 (Conference on Lifelong Learning Agents) Web Communication Chair	Nov 2025 - Current
Managing the official conference website and digital presence for the 2026 edition.	
MILA Teaching Assistant – Summer High School Internships	Summer 2025
Designed and delivered educational workshops on Machine Learning and Probability for high school students.	
UILO – Career Orientation Platform Co-Creator and Technical Lead	Sept 2024 – Present
Built UILO, a career orientation platform for post-secondary education with university and career databases and psychometry-based recommendations.	
Belle Parole Arci APS Co-Founder, Non-Profit Association	Nov 2023 - Current
Organized community events and communications, built the organization’s website belleparoleaps.it, and secured grant funding for organizational activities.	

ADDITIONAL SKILLS, SERVICE, AND ACHIEVEMENTS

Reviewing	CoLLAs 2026; MemAgent Workshop at ICLR 2026.
Skills	PyTorch, Hugging Face, vLLM, LangChain/LangGraph, Python, Linux, L ^A T _E X, Weights & Biases, Figma.
Recognition	Selected Participant, Y Combinator AI Startup School (Jun 2025).
Languages	English (professional proficiency); Italian (native proficiency).